

# RT8V-M-BD

**108 Cells**

Topcon Mono 16 BB

**490-515W**

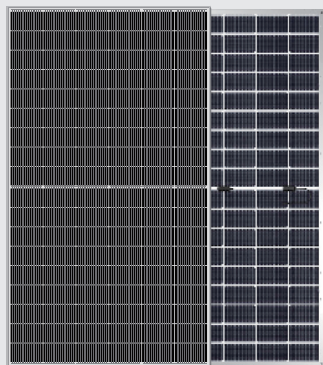
Power output

**23.16%**

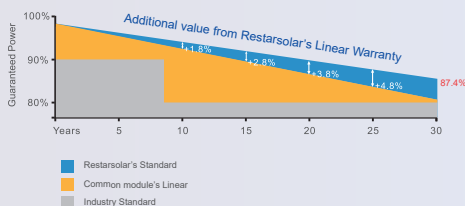
The Highest Efficiency

**0~+5W**

Tolerance



**0.42% Annual Degradation  
over 30 years**



LINEAR PERFORMANCE WARRANTY

**15 Year Product Warranty**

**30 Year Linear Power Warranty**

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## RT8V-M-BD(TOPCON)

RT8V-M-BD HALF-CELL series is produced with high efficiency multi-busbar cells, which can reduce the module internal power loss to improve its conversion efficiency, as well as lower the failure risk caused by cracks and broken busbar to enhance the module reliability. Combined with half-cell technology, the module is highly resistant to hot-spot crisis caused by shadow effect.



### High Reliability

Multi-busbar technology can effectively reduce the reliability risk caused by cells cracks and broken busbar.



### Anti-PID Resistance

Prominent anti-PID performance reduces the power degradation, leading to higher energy yield and lower LCOE.



### Durability Against Extreme Conditions

Certified to resist high salt mist and ammonia conditions.



### High Efficiency

Multi-busbar technology can reduce the module internal power loss to improve the module conversion efficiency significantly.



### Low-Light Performance

With high transmittance and anti-reflective 3.2mm tempered glass, the module has stronger performance under low light circumstances.



### High Mechanical Strength

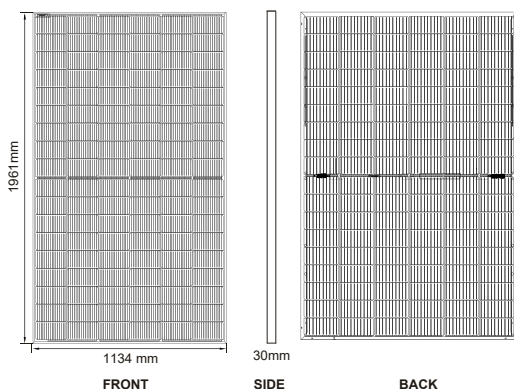
Certified to withstand: high wind load(2400Pa) and snow load(5400Pa).

## Full range of products and certification systems

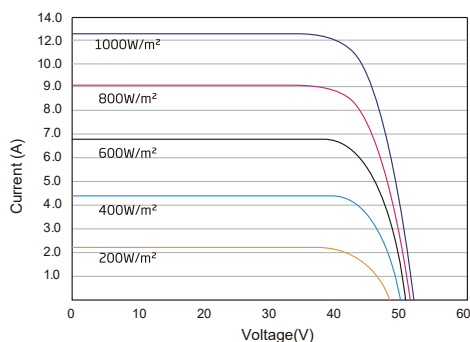
ISO 9001/14001 TUV PID-FREE CE IEC 61215/61730/61701/62716



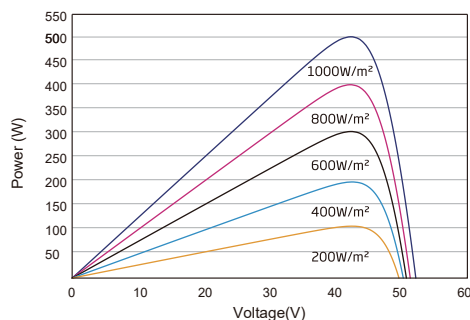
Dimension of PV Modules Unit: mm



I-V CURVES OF PV MODULE(500 W)



P-V CURVES OF PV MODULE(500W)



## ELECTRICAL DATA(STC)

|                               |        |        |        |        |        |        |
|-------------------------------|--------|--------|--------|--------|--------|--------|
| Rated Power in Watts-Pmax(Wp) | 490    | 495    | 500    | 505    | 510    | 515    |
| Open Circuit Voltage-Voc(V)   | 39.15  | 39.35  | 39.55  | 39.75  | 39.95  | 40.15  |
| Short Circuit Current-Isc(A)  | 15.94  | 16.00  | 16.06  | 16.12  | 16.18  | 16.24  |
| Maximum Power Voltage-Vmp(V)  | 32.65  | 32.85  | 33.05  | 33.25  | 33.45  | 33.65  |
| Maximum Power Current-Imp(A)  | 15.01  | 15.07  | 15.13  | 15.19  | 15.25  | 15.31  |
| Module Efficiency(%)          | 22.03% | 22.26% | 22.48% | 22.71% | 22.93% | 23.16% |

STC: Irradiance 1000 W/m², Cell Temperature 25°C, Air Mass AM1.5 according to EN 60904-3.

## ELECTRICAL DATA(NOCT)

|                              |       |       |       |       |       |       |
|------------------------------|-------|-------|-------|-------|-------|-------|
| Maximum Power-Pmax(Wp)       | 370   | 374   | 378   | 382   | 386   | 389   |
| Open Circuit Voltage-Voc (V) | 36.68 | 36.87 | 37.06 | 37.25 | 37.43 | 37.62 |
| Short Circuit Current-Isc(A) | 12.88 | 12.93 | 12.98 | 13.02 | 13.07 | 13.12 |
| Maximum Power Voltage-Vmp(V) | 30.63 | 30.81 | 31.00 | 31.19 | 31.38 | 31.56 |
| Maximum Power Current-Imp(A) | 12.10 | 12.14 | 12.19 | 12.24 | 12.29 | 12.34 |

NOCT: Irradiance at 800 W/m², Ambient Temperature 20°C, Wind Speed 1m/s.

## INTEGRATED PWER @ STC (REFERENCE TO 10%)

|                               |        |        |        |        |        |        |
|-------------------------------|--------|--------|--------|--------|--------|--------|
| Rated Power in Watts-Pmax(Wp) | 539    | 545    | 550    | 556    | 561    | 567    |
| Open Circuit Voltage-Voc(V)   | 39.15  | 39.35  | 39.55  | 39.75  | 39.95  | 40.15  |
| Short Circuit Current-Isc(A)  | 17.50  | 17.58  | 17.63  | 17.70  | 17.75  | 17.83  |
| Maximum Power Voltage-Vmp(V)  | 32.65  | 32.85  | 33.05  | 33.25  | 33.45  | 33.65  |
| Maximum Power Current-Imp(A)  | 16.51  | 16.59  | 16.64  | 16.72  | 16.77  | 16.85  |
| Module Efficiency(%)          | 24.24% | 24.51% | 24.73% | 25.00% | 25.23% | 25.50% |
| BNPI/Isc(BSI)                 | 19.53  | 19.60  | 19.67  | 19.75  | 19.82  | 19.89  |

## MECHANICAL DATA

|                    |                                        |
|--------------------|----------------------------------------|
| Solar Cells        | Mono-crystalline 182*105mm, 16Bus bars |
| Cell Configuration | 108cells(6*18)                         |
| Module Dimensions  | 1961*1134*30mm                         |
| Weight             | 22.7kg                                 |
| Front Cover        | 3.2mm Tempered Glass                   |
| J-Box              | IP68                                   |
| Cable              | 4mm²(IEC)/12AWG(UL), 300mm+300mm       |
| Connectors         | MC4 or MC4 Comparable                  |
| Standard Packaging | 36/37pcs/pallet                        |

## TEMPERATURE & MAXIMUM RATINGS

|                                          |                      |
|------------------------------------------|----------------------|
| Nominal Operating Cell Temperature(NOCT) | 45°C±2°C             |
| Temperature Coefficient of Voc           | -0.25%/°C            |
| Temperature Coefficient of Isc           | 0.04%/°C             |
| Temperature Coefficient of Pmax          | -0.29%/°C            |
| Operational Temperature                  | -40~+85°C            |
| Maximum System Voltage                   | 1500V(IEC)/1500V(UL) |
| Max Series Fuse Rating                   | 25A                  |
| Limiting Reverse Current                 | 25A                  |

## PACKAGING CONFIGURATION

|                                 |                  |
|---------------------------------|------------------|
|                                 | 40HQ             |
| Number of modules per container | 864/888 pcs      |
| Package                         | 36/37 pcs/pallet |
| Package Number                  | 24 pallets       |